



The Merit-Order Effect on the Swedish bidding zone with the highest electricity flow in the *Elspot* market

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ABSTRACT

Electricity interconnections have become increasingly common as a means of integrating electricity markets, expediting the exchange of electricity, and thereby creating balanced electricity systems. These interconnections also facilitate the incorporation of intermittent renewable energy sources, ensuring greater energy security and reliability in electricity markets. In this study, an assessment is made of the impact of both electricity produced from wind power, and the cross-border flow of electricity, on the mean and volatility of the day-ahead electricity price. To do this, we studied the Swedish bidding zone with the highest flow of electricity imports and exports between 2016 and 2019 – the SE3 bidding zone (BZ). Using a SARMAX/GARCH approach, every hour of the day was assessed individually, by estimating up to 24 different models from 1 January 2016 to 30 April 2020. The results confirm the Merit-Order Effect from wind power, and that its impact is similar in magnitude throughout the day. Furthermore, the flow of electricity from both import and export appear to increase the day-ahead electricity price. This study also suggests that the volatility of electricity prices is primarily increased by unexpected shocks. This may be related to the high dependence of the SE3 BZ electricity system on electricity imports, which makes the system extremely susceptible to larger transmission of shocks.

1. Introduction

Electricity markets worldwide are undergoing a period of transition towards sustainable electricity systems. The integration of new renewable energy sources (RES) such as wind and solar photovoltaic (PV) has intensified volatility in electricity prices in most electricity markets (e.g., Sapio (2019), Maciejowska (2020) and Rai and Nunn (2020)). RES production is highly intermittent and depends on weather conditions that can change unexpectedly. This can lead to a mismatch between electricity demand and electricity supply from RES, and this has been also blamed for the non-linearity observed in electricity prices. Thus, electricity systems are prone to be highly unstable, so that they require greater flexible backup generation to ensure a steady supply of electricity. However, it is claimed that electricity produced from RES is reducing the day-ahead electricity price, due to their nearly zero marginal costs. This phenomenon, also known as the merit-order effect (MOE), may render flexible generators less profitable, thus decreasing the incentive to invest in backup capacity that is crucial for ensuring the

reliability of electricity systems.

Since the capacity for storing electricity on a large scale is currently close to nil, many electricity markets have installed interconnections to share the surplus (and mitigate the scarcity) of RES electricity, thereby reducing price differences between the connected countries, and balancing the overall system. However, unless there is optimal management of interconnections, the intensified electricity flow may expose integrated electricity markets to pricing risks. For instance, in an integrated electricity market, a sharp shock in electricity supply or demand in a specific bidding zone (BZ), may also have an impact in neighbouring BZs. Briefly, the integration of wind and solar PV power into electricity systems brings up several fundamental issues. For instance, how investors in energy sources perceive the risk of electricity pricing, and the extent to which the marginal pricing principle will incentivize future investments in additional flexible generation capacity. These issues need to be taken into account when devising strategies to encourage the incorporation of intermittent RES, while fostering solid and resilient electricity market infrastructures.

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The aim of this research is to assess the impact of the flow of electricity imports and exports (hereafter, inflow and outflow, respectively), as well as the MOE from wind power on the day-ahead electricity price. Between 2016 and 2019, the Swedish BZ with the highest outflow and inflow among the countries of the Nord Pool electricity market, was the SE3 BZ, so this was the zone analysed. Pronounced volatility in electricity prices can jeopardise the reliability of a transmission grid. Given this, it is crucial for both investors and regulators to understand the behaviour of electricity prices so that the transition to greener electricity systems can be dealt more efficiently. Therefore, this study also assesses volatility in the time series of electricity prices in the SE3 BZ. It covers the period from 1 January 2016 to 30 April 2020 (using all the available data). The approach used involved a Seasonal Autoregressive Moving Average model with exogenous regressors (SARMAX), combined with a Generalized Autoregressive Conditional Heteroskedasticity model (GARCH).

The impact on electricity prices of both wind power generation and electricity consumption, may exhibit periodicity. For example, it may depend on the hour-of-the-day, or day-of-the-week, and in turn, this may trigger high seasonality in electricity price time series. Hence, the volatility of electricity prices may differ, depending on the balance between the instantaneous supply and demand of electricity. To consider this issue, each hour of the day was analysed by estimating up to 24 different SARMAX/GARCH models. This study intends to answer the following research questions: (i) does the MOE from wind power occur at all hours of the day?; (ii) does the volatility of electricity prices in the SE3 BZ increase with rises in electricity production from wind power?; (iii) what is the impact of the inflow and outflow of electricity on the day-ahead electricity price in the SE3 BZ?; and (iv) is the volatility of the electricity price more vulnerable to new shocks, or to past volatility?

The main contribution of this paper to the literature lies in its simultaneous analysis of the MOE, and of the impact of cross-border flows on the mean and variance of the day-ahead electricity price. Electricity flow is likely to have a particularly strong influence on the formation of domestic electricity prices, particularly in the SE3 BZ, where nuclear power represents the largest share of electricity production. Finally, this paper also provides a more detailed analysis of the day-ahead electricity price, by considering each hour separately, thereby making a further valuable contribution to the literature, and enabling policymakers to draw appropriate conclusions and devise rules tailored to the dynamics observed in peak and off-peak hours of consumption.

This paper is set out as follows: Section 2 presents a literature review; Section 3 briefly describes the Swedish electricity system; Section 4 analyses the data; Section 5 describes the methodology; Section 6 presents and discusses the results; and Section 7 presents the conclusions.

2. Literature review

Rising concerns about environmental degradation have led to a profound redesign and restructuring of electricity markets worldwide. The idiosyncratic characteristics of RES, such as intermittency and negligible marginal costs (particularly of wind and solar PV power), have been challenging electricity systems, with unforeseen consequences. These include a decrease in electricity prices; an increase in electricity price volatility; and increased demand for electricity generated from flexible and/or controllable energy sources to back up RES generation and guarantee reliable electricity supply, which usually have higher marginal costs.

The impact of electricity generation from wind and solar PV power on electricity prices has been the subject of research on various electricity systems. The finding that electricity prices decrease with increased production of electricity by RES (i.e. the MOE) remains consensual in most studies (e.g. (Würzburg et al., 2013), (Cludius et al., 2014), (Ketterer, 2014), (Woo et al., 2016), and (Cserekyei et al., 2019)). However, the magnitude of the MOE may vary according to the specific features of the electricity generation mix of each electricity

market. In Slovakia, the MOE from solar PV power was found to be negligible (Janda, 2018), but this result is not unreasonable, since nuclear power plays a prominent role in the generation mix there, and the number of sunny days in this country is also low. Similarly, in the case of the Czech Republic, the MOE from solar PV source was not marked, although it was also found for wind and hydro energy sources (albeit with low magnitude) (Luňáčková et al., 2017). Despite the substantial installed capacity of solar PV power in the Czech Republic, weather conditions in the country limit its potential. Meanwhile, in Italy, the MOE from wind and solar power has been decreasing over time, as a result of an increase in electricity produced from intermittent RES (Clò et al., 2015).

The Nord Pool electricity market has a large installed capacity of interconnections, which give it ample opportunity to share backup capacity between its cross-border member countries. Unger et al. (2017) found the MOE to be significant in the electricity systems of Denmark and Finland, namely, in the DK1 and FI BZs, respectively. The authors concluded that, on the one hand, interconnectivity may cause a decrease in electricity prices, but on the other hand, could make other electricity systems sensitive to high electricity prices, especially when a country is net importer. Interconnectivity can help to efficiently manage the electricity production and consumption of an integrated electricity market. Electricity interconnections may reduce the need to install idle capacity (usually expensive flexible generation), which would only be used to guarantee self-sufficiency and satisfy peaks of consumption. However, these interconnections also expose electricity markets to a much more fluid propagation of risk, as a shock in one country may spread more easily to all the countries in the connected market. This phenomenon was observed in the Nord Pool electricity market, where net exporters are (generally) the spreaders of volatility (Sorknæs et al., 2019). A study made of other European power markets highlighted that France, Ireland and Portugal have the most volatile electricity systems, while major changes in volatility have a larger impact, for a (relatively) longer period in future volatility in Nordic countries, Ireland, and the United Kingdom (Erdogdu, 2016).

The German electricity system is one of the leading systems in the development of RES, so it has been the subject of extensive research. Electricity generated from wind power seems to increase the frequency and duration of episodes of low electricity prices (Lagarde and Lantz, 2018). Furthermore, electricity production from wind power also seems to increase the volatility of electricity prices in Germany. Conversely, solar PV power reduces this volatility by averting the need to use power plants with high marginal costs during peak loads (Kyritsis et al., 2017). In Denmark, by contrast, electricity price volatility is low when there are high amounts of electricity produced from wind power (Rintamäki et al., 2017). The authors suggest that this could be related to patterns of wind power generation that produce electricity at different times of the day, making the volatility of Danish electricity price less sensitive to the downward effect of RES on electricity prices.

The Iberian electricity market is composed of the Portuguese and Spanish electricity systems, whose habits and periods of electricity consumption are quite similar. Despite this, the interaction between electricity sources in each of these countries were found to be different (Afonso et al., 2019). The electricity systems of Portugal and Spain are different in scale, for example, backup capacity in Spain is not comparable to that in Portugal. Electricity production from wind power in Portugal appears to increase the volatility of wholesale electricity price and the leverage effect was also confirmed (Macedo et al., 2020). Wind power production in Portugal occurs predominantly during off-peak hours, and this could create high volatility in electricity prices. A negative relationship between variable RES and electricity prices was also found in Spain (Ballester and Furió, 2015). The high variability of electricity produced from RES is likely to induce volatility in the price of electricity. In contrast, combined-cycle electricity production seems to reduce it. In addition, uncertainty in Spanish energy regulation is also considered by Ciarreta et al. (2020) to be one of the factors inducing

high instability in electricity prices. Market-based measures would encourage a more efficient integration of intermittent RES in electricity markets.

In the context of Portugal and Spain, the MOE was calculated by Figueiredo and da Silva (2019), and it appears to be positively influenced by electricity consumption, and wind and solar power. Meanwhile residual load (or net demand) seems to have an adverse effect on the MOE. The authors suggest that increases in production from RES could avoid power plants with high marginal costs setting the wholesale electricity price. Focusing on the German electricity market, Pereira and Marques (2020a, 2020b) found that net demand has a higher capacity to explain the formation of wholesale electricity prices than traditional electricity demand.

The implications of over- and under-forecasting wind power production were assessed in the New-England power system by Martinez-anido et al. (2016). The authors suggest that over-forecasting wind production increases electricity prices, while under-forecasting decreases them. Gürtler and Paulsen (2018) proved that avoiding forecast errors for electricity production from solar PV and wind power is important for decreasing volatility in electricity prices. Moreover, smoothing electricity demand would also contribute to this reduction.

The transition of electricity markets towards green energy systems, also implies the decommissioning of conventional power plants. Increased electricity consumption during peak hours in France seems to be hastening the phasing-out of nuclear plants, due to the urgent need to develop flexible generation capacity (Pereira and Marques, 2020b). Phasing out nuclear plants could lead to reduced security of electricity

supply, and/or potential increases in electricity prices, as noted by Kolb et al. (2020). The authors also highlight that the installed capacity of RES must be increased to overcome this issue, particularly as RES production has been shown to contribute significantly to the security of supply in Germany. Another notable effect of the MOE on electricity systems is a decline in the market value of wind and solar PV (Zipp, 2017). The forthcoming adaptation of electricity markets to new designs between 2025 and 2040 in Spain was assessed by Gerres et al. (2019). The authors highlight the pressing need for remuneration mechanisms to enable RES to achieve a sufficiently comfortable market share, and guarantee security of supply. However, these mechanisms must acknowledge uncertainties in the electricity market and, thus, provide additional hedging strategies to guarantee a certain revenue for market participants.

In summary, time series of electricity prices have been analysed in several studies, for different electricity systems. However, no recent literature has analysed each hour of the day individually, or the use of cross-border electricity flow to explain either the volatility or the mean of electricity prices. As the integration of electricity markets becomes increasingly common (often by boosting the installed capacity of electricity interconnections), a study of its implications is extremely important and timely. Consequently, this research intends to provide an analysis of the dynamics of an electricity system with substantial imports and exports of electricity, by analysing each hour of the day individually.

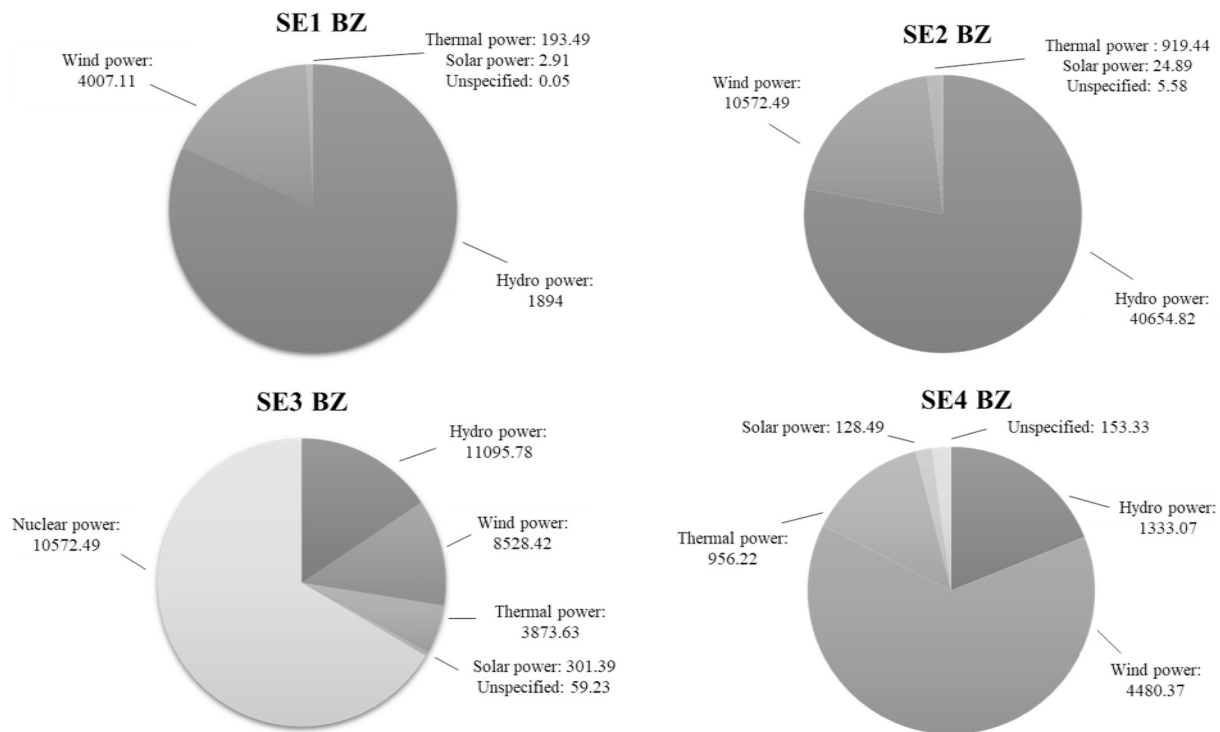


Fig. 1. Total of electricity generation (GWh) in each BZ of the Swedish Electricity system in 2020.

Note: Data Source: Svenska kraftnät authority. Graphs by the authors.

3. Brief debate on the dynamics of the Swedish electricity system

The Swedish electricity system is included in the Nord Pool electricity market which is one of the most integrated electricity markets worldwide. Electricity generation in Sweden is predominantly from hydro and nuclear power, although wind power production has been increasing significantly. The electricity system in Sweden has four BZs (i.e., SE1, SE2, SE3 and SE4), each with generation structures of different dimensions. The electricity generation mix of each BZ and the total electricity produced in 2020 is shown in Fig. 1.

According to information published by the Svenska kraftnät authority, the BZs of northern Sweden (SE1 and SE2) usually have an electricity consumption deficit, which means that electricity production is higher than electricity consumption. Meanwhile, in the southern BZs (SE3 and SE4), electricity consumption is higher than electricity production levels, meaning that these BZs tend to import electricity. Therefore, the southern BZs are likely to be net importers, which makes the assessment of the impact of electricity flow on electricity prices in these BZs of particular interest.

Fig. 2 shows the hourly average of electricity flow of each BZ of the Nord Pool electricity market in each year from 2016 to 2019, which was calculated as follows:

$$\frac{1}{N} \sum_i (\text{inflow}_i + \text{outflow}_i), \quad (1)$$

where N represents the total number of hours in a year and i refers to the BZ.

This paper focuses on studying the SE3 BZ, whose electricity production is predominantly from nuclear power (cf. Fig. 1). Given the low marginal costs of nuclear generation, one would expect the MOE of wind power to be smaller here than in other electricity systems with more expensive backup electricity generation. Besides this, the SE3 BZ had the highest hourly average of electricity flow within the Nord Pool electricity market in every year from 2016 until 2019 (cf. Fig. 2). The exchange of electricity between BZs is likely to minimize imbalances in electricity systems and thus, increase the overall flexibility of integrated electricity markets. Thus, the idiosyncratic features of the SE3 BZ, i.e., its electricity production deficit, high electricity flow and the prominent role of nuclear power, led to choose this BZ to study of the dynamics of electricity prices.

4. Data analysis

This paper analyses the data on an hour-by-hour basis, i.e., by estimating up to 24 independent models. This kind of analysis, also known as a multivariate modelling framework (Ziel and Weron, 2018), assesses each hour individually, and potential interdependencies between explanatory variables across hours are not considered. Since each hour is individually analysed, there is a 24-h lag between variables and thus, bids from day d are not submitted at the same time as bids from day $d - 1$, for instance. Consequently, the well-known autocorrelation of hour h of day d with respect to hour h of day $d - 1, \dots, d - p_h$ are substantially reduced.

The data was obtained from the Nord Pool official website, from 1 January 2016 to 30 April 2020, for the SE3 bidding zone. The sample period was chosen based on data availability, and the total number of observations considered in each estimation was 1582. The variables

(and their units) included in the database were: the Elspot electricity price ($PRICE$, €/MWh); the prognosis of electricity consumption ($CONS$, MWh); the prognosis of electricity produced from wind power ($WIND$, MWh); electricity inflow ($INFLOW$, MWh); and electricity outflow ($OUTFLOW$, MWh). All variables were transformed into logarithms to reduce non-normality and/or eventual noise in the data. It is worthwhile noting that, in some hours of the day, the day-ahead electricity price displays a negative value, which does not exceed -1 €/MWh . Therefore, to make the logarithmic transformation feasible, we have added a constant of one to all variables, as suggested by Lagarde and Lantz (2018).

Fig. 1 shows the logarithm of electricity price series for each hour of the day.

From a visual inspection of the electricity price graphs (Fig. 3) there is a discernible increase in price variations between 6 a.m. and 6 p.m. This phenomenon may result from the greater dynamics of the electricity market during this period, which are probably due to increased electricity consumption (cf. Fig. 4). The shapes of the time series observed in all of the 24-h electricity price graphs demonstrate similar annual seasonality. Furthermore, downward spikes were more marked at night and in the early morning (i.e., between 10 p.m. and 6 a.m.), when peak wind production and off-peak electricity consumption are likely to occur. Furthermore, it is also clear that electricity prices varied significantly in the first months of 2020, throughout the 24-h period, which could be a collateral effect of the Covid-19 pandemic.

Electricity prices in Sweden may be influenced by several factors, in which both electricity demand and production play a crucial role. The Swedish residential sector has a high electricity demand, mostly for space heating. Therefore, peak electricity consumption could be a key factor in price setting. However, the setting of day-ahead electricity prices is not just limited to the inherent features of a specific market area (its daily habits and/or electricity production), but is also affected by those observed in other connected countries, particularly where there is higher interconnection capacity and closer market coupling. For instance, the simultaneous combination of high electricity production from wind power, with low electricity consumption in the SE2 BZ, would not only have a large downward effect on electricity prices in this area, but also in neighbouring connected areas (e.g. the NO1, DK1, and FI). In view of the high interconnectivity between Nordic Baltic countries, the inclusion of electricity flow in econometric models seems extremely relevant for assessing day-ahead electricity prices. Thus, the trade in electricity within these countries has the potential to worsen and/or alleviate increased residential demand, market failures and system imbalances.

4.1. Descriptive statistics

Table A.1, in the appendix, shows the descriptive statistics of the data for each of the 24 h of the day. The results revealed that none of the variables are normally distributed. The distribution of $LPRICE$ showed significant negative skewness in all 24 h. This suggested that the day-ahead electricity price is highly skewed, regardless of the hour of the day. Accordingly, the Swedish day-ahead electricity price exhibited a fat left tail, and the occurrence of electricity prices below their mean were more frequent (i.e., downward jumps) than those above it. A priori, this may be consistent with the negative effect of electricity production from variable RES on electricity prices. The distributions of $LWIND$ and $LCONS$ also exhibited negative skewed values, albeit not as significant as those observed in $LPRICE$ (the distributions of $LCONS$ at 11 p.m. and 12 p.m. actually showed positive skewed values). With respect to the

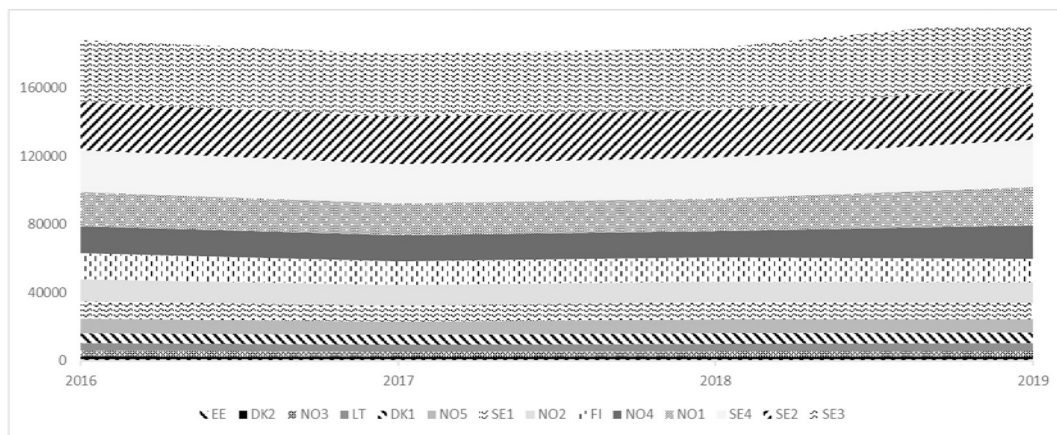


Fig. 2. Hourly average of the electricity flow for each BZ in the *Elspot* market from 2016 to 2020^a.

Note: Data source: Nord Pool Official Website. Graph by the authors.

^aThe meaning of the acronyms of the BZs is as follows: NO1, NO2, NO3, NO4 and NO5 refer to Norway's BZs; SE1, SE2, SE3 and SE4 denote Sweden's BZs; DK1 and DK2 stand for Denmark's BZs; FI is the Finnish BZ; EE represents the Estonian BZ; and LT indicates the Lithuanian BZ. In order to make them more readable, in the legend the BZs are represented in reverse order compared to the graph.

LINFLOW and *LOUTFLOW* variables, the results exhibited slightly higher negative skewed values, mainly at night and early morning, than the abovementioned variables.

As expected, there was strong evidence for high kurtosis for *LPRICE* in each of the 24 h of the day. *LOUTFLOW* was the most leptokurtic variable during the night (12 p.m. and 4 a.m.), while the *LINFLOW* series played this role between 5 a.m. and 9 a.m. This finding is reasonable, since extreme electricity outflow and inflow occurrences are supposed to occur during the off-peak and peak demand periods, respectively. These results appeared to be in line with the management goals of balancing the electricity system. In short, given the features of the distribution of Swedish electricity prices (i.e., significant negative skewness and high kurtosis), the Student-*t* distribution seemed a valid assumption to consider in model estimations. Using the *t*-location scale distribution, the appropriate location (μ), scale (σ), and shape (ν) parameters were established to better fit the models. This adjustment in the parameters of the Student-*t* distribution, according to the statistical idiosyncrasies of the data, take into account the negative skewness and high kurtosis usually displayed in the day-ahead electricity price.

4.2. Unit root tests

The stationarity proprieties of variables were assessed using the Dickey-Fuller Generalized Least Squares (DF-GLS) unit root test (Elliott et al., 1996). The DF-GLS is an alternative to the traditional Augmented Dickey Fuller unit root test of Dickey and Fuller (1979), which has been shown to be substantially more effective in assessing the stationarity of time series with undetermined average and/or tendency. Under the null hypothesis of the presence of a unit root, the DF-GLS test proves that variables are stationary in most hours. Nonetheless, it was also found that, at a few hours, the day-ahead electricity price time series appears be non-stationary.

Hourly time series with high persistence, peaks, time-varying and potential heteroskedasticity (e.g., day-ahead electricity price time series) may indeed possess long-memory, if episodes of persistence (or correlation) are frequent for many periods. Fractional integrated processes are often applied to disclose such correlation structures. Chang and Perron (2017) argue that standard unit root test usually reject the null hypothesis when the exact process of a variable is fractionally integrated. This may lead to biased conclusions regarding the stationarity of time-series variables and therefore, the multivariate semiparametric estimate of long memory of Robinson (1995) was performed.

The null hypothesis of Robinson (1995) i.e., $H_0 : d = 0$, assesses whether a time series variable exhibits short-memory, which means stationarity and invertible ARMA modelling. Table A.2 in the appendix presents the estimates of fractional differencing parameter resulting from the Robinson's test, and suggests that all variables tend towards a high degree close to one, in all 24 series/h. This may indicate that variables exhibit long-memory (and/or high persistence), and thus, are fractionally integrated.

To identify possible structural breaks in the electricity price time series, the Bai and Perron (2003) test was also applied. The Bai and Perron (2003) test performed in this study employed global maximizes to determine breakpoints (where serial correlation and/or heteroskedasticity is allowed), in which error distributions are permitted to differ across breaks. The null hypothesis is no breaks, against the alternative hypothesis of l breaks, and in this case the maximum number of breaks tested was five. The Bai and Perron (2003) multiple breakpoint test was calculated for all variables.¹ The results indicated the presence of structural breaks in the day-ahead electricity price series, in all of the 24 h. Overall, Swedish electricity prices seem to not display strong patterns of volatility clustering (cf. Fig. 4), so that the structural breaks detected by both the Bai and Perron (2003) and Zivot and Andrews (1992) tests are actually prominent outliers. Please note that most of these outliers were corrected in the estimation of the models by impulse and shift dummies, which is one of the techniques suggested by de Marcos et al. (2020), and that the inclusion of dummies in estimations to deal with structural breaks was a procedure also followed by Loi and Le Ng (2018).

5. Method

The integration of intermittent RES in electricity markets has increased the exposure of electricity systems to new risks and vulnerabilities. The most stressful effects, challenging the reliability and security of electricity supply systems, have been increased volatility, and decreases in electricity prices. Electricity price series usually exhibit autocorrelation and are highly heteroskedastic. Therefore, when assessing electricity prices, it is important to take into account their

¹ Please note that to save space, the results of the DF-GLS and the Bai and Perron (2003) multiple breakpoint test are not presented, but they are available upon request to the authors.

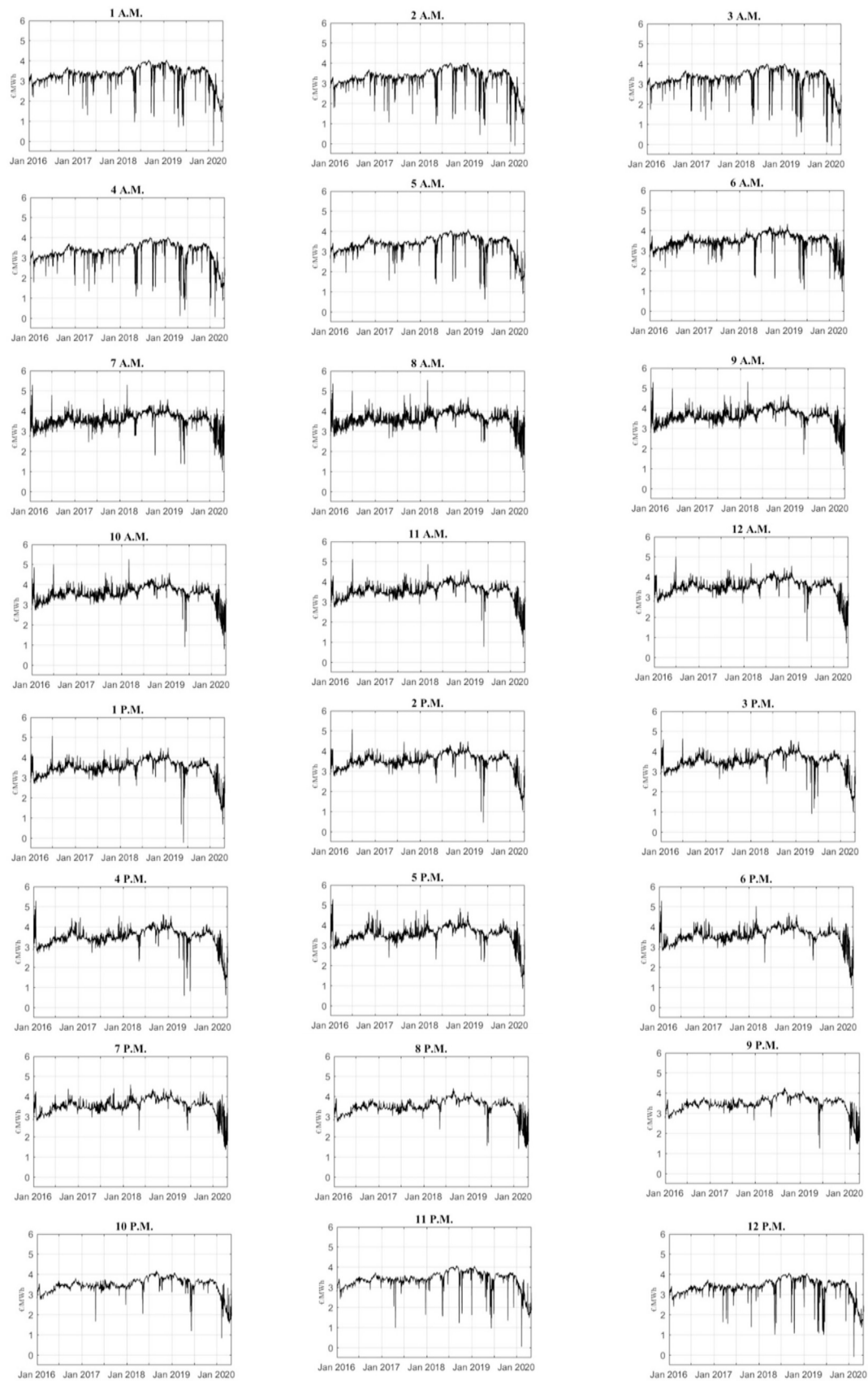


Fig. 3. Logarithmic electricity price series for each hour of the day in the SE3 bidding zone.
Note: Data source: Nord Pool Official Website. Graph by the authors

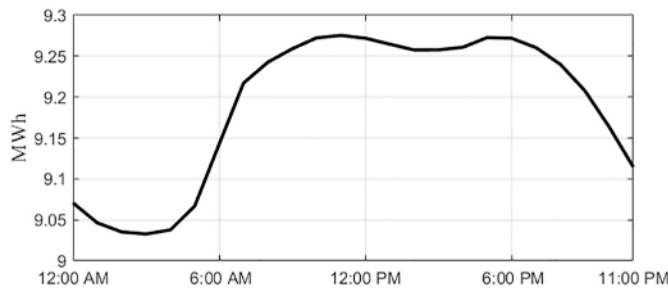


Fig. 4. Sample average of the logarithm of the prognosis of electricity consumption across the day, from 1 January 2016 to 30 April 2020.

Note: Graph by the authors.

inherent features, such as volatility-clustering, mean-reversion, outliers, and seasonality. Because of its potential to explain most of these characteristics, the methodology applied in this research was the SARMAX(p, q)(P, Q)/GARCH(p, q).

Given the current inability to store electricity on a large scale, RES production cycles are still not synchronized with electricity consumption cycles. This mismatch is often manifested in unanticipated spikes that may reflect simultaneous periods of low consumption and excess electricity production from RES; and/or, periods of high demand for flexible generation due to low levels of production from RES. Because these commonly-observed extremes in the day-ahead electricity price should be considered while studying the MOE, the outliers in the electricity price time series were not removed. Nevertheless, a set of both of impulse and shift dummies were included in estimations to rectify spurious observations.

5.1. GARCH (p, q) models

The GARCH model was developed by Bollerslev (1986), and it is commonly used in time series in the presence of high, time-varying volatility. The general structure of this approach can be written as:

$$P_t = \omega + \delta X_t + \epsilon_t, \quad (2)$$

$$\epsilon_t = \sigma_t z_t \quad (3)$$

$$\sigma_t^2 = \gamma + \sum_{i=1}^p \beta_i \sigma_{t-i}^2 + \sum_{j=1}^q \alpha_j \epsilon_{t-j}^2, \quad (4)$$

where, P_t represents the Swedish day-ahead electricity price; X_t depicts the contemporaneous exogenous variables; ϵ_t denotes the error term of σ_t^2 ; z_t describes a set of iid variables with a distribution of zero mean and variance one; ω and γ are the constant terms; δ , β_i and α_j stand for the parameter coefficients; and p and q are the orders of the GARCH and ARCH parameters, respectively. The stationarity of GARCH estimations is underlined when the following conditions are met: (i) the roots of $\beta(z) = 1 - \beta_1 z + \dots + \beta_p z^p$ and $\alpha(z) = 1 + \alpha_1 z + \dots + \alpha_q z^q$ must be placed outside the unit circle; and (ii) $\alpha_j + \beta_i < 1$ must be verified for every j, i such that $j = 1, \dots, p$ and $i = 1, \dots, q$, whereas $\alpha_j, \beta_i > 0$ so that the unconditional volatility is strictly positive. The non-stationarity condition of GARCH models is true when $\alpha_j + \beta_i \geq 1$.

The α_j , known as the GARCH error parameter (or ARCH parameter), evaluates the impact of new shocks (regardless of their direction) in a time series. Meanwhile, the impact of past shocks on future volatility is assessed by β_i , called the GARCH lag parameter. Overall, the persistence of volatility is measured by the results obtained from $\alpha_j + \beta_i$. In other words, volatility is found to be highly persistent when $\alpha_j + \beta_i \cong 1$, whereas, if $\alpha_j + \beta_i \ll 1$ the volatility's persistence is significantly low. The high persistence of volatility suggests that a shock in a time series is likely to revert slowly, whereas the low persistence of volatility suggests spikes will tend to die away quickly.

5.2. SARMAX(p, q)(P, Q)/GARCH(p, q)

5.2.1. Mean equation

The specification of the mean equation in this research incorporates the Seasonal Autoregressive Moving Average with exogenous regressors (SARMAX) structure of Box and Jenkins (1976). An analysis of the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots (correlograms) led to the specification of AR(p) and MA(q) parameters in the mean equation. Furthermore, the strong pattern of seasonality in the time series of day-ahead electricity prices (previously identified in Fig. 3) was also clear in the ACF and PACF plots. This prompted the inclusion of both seasonal autoregressive (SAR(P)) and moving average (SMA(Q)) parameters in the mean equation.²

A detailed description of the mean equation of SARMAX(p, q)(P, Q)/GARCH(p, q) can be found in Papaioannou et al. (2018), and its structure for each hour of the day can be written as follows:

$$P_{t,h} = \omega + \sum_{j=1}^{p_h} \phi_j P_{t-j} + \sum_{i=1}^{P_h} \Phi_i P_{t-s_h} - \sum_{i=1}^{P_h} \Phi_i \sum_{j=1}^{p_h} \phi_j P_{t-j-s_h} + \sum_{j=1}^{q_h} \varphi_j \epsilon_{t-j} + \sum_{z=1}^{Q_h} \psi_z \epsilon_{t-s_h} + \sum_{z=1}^{Q_h} \psi_z \sum_{j=1}^{q_h} \varphi_j \epsilon_{t-j-s_h} + \sum_{l \in k} \delta_l X_{t,l} + \epsilon_t, \quad (5)$$

where, ϕ_p and Φ_P stand for the non-seasonal and seasonal AR parameters, respectively; φ_q and ψ_Q denote the non-seasonal and seasonal MA parameters; s represents the seasonal term of both the SAR and SMA processes; h means hour; k signifies the information set; δ is the coefficient of the contemporaneous exogenous variables; and l and X indicate the summation index and the vector of the contemporaneous exogenous variables, respectively.

5.2.2. Variance equation

The equation of conditional variance in the SARMAX(p, q)(P, Q)/GARCH(p, q) approach is represented by the following expression:

$$\sigma_t^2 = \gamma + \beta_1 \sigma_{t-1}^2 + \alpha_1 \epsilon_{t-1}^2 + \sum_{m \in r} \lambda_m Z_{t,m}, \quad (6)$$

where, λ and Z represents the coefficient of contemporaneous exogenous variables and their vector, respectively; m denotes the information set, and r the summation index. The exogenous regressors considered in both mean and variance equations are: the prognosis of electricity generation from wind power; the prognosis of electricity consumption; and both electricity inflow and electricity outflow.

6. Results and discussion

6.1. Results

The volatility of the Swedish day-ahead electricity price, as well as the MOE, were assessed hour-by-hour, by modelling up to 24 different SARMAX/GARCH models. The data frequency was daily, and the period from 1 January 2016 to 30 April 2020 was analysed. The maximum likelihood estimator and the Student- t distribution were the options selected in estimating all the model. The empirical results are reported in Tables 2 and 3, and are interpreted in terms of elasticities (as all variables are measured in logarithms). In order to simplify the interpretation of results, the hours starting from 7 a.m. to 7 p.m. were considered to be the peak, and the hours starting from 8 p.m. to 6 a.m. to be the off-peak periods of consumption.

The diagnostic tests used to check the goodness of fit and accuracy of SARMAX/GARCH estimates were the Ljung-Box Q-Statistics and the

² Please note that, to preserve space, the ACF and PACF plots of each of the 24-time series of day-ahead electricity price are not presented, but they are available, upon request to the authors.

Table 1
Results of SARMAX/GARCH models between 1 a.m. and 12 a.m.

Variables	1 a.m.	2 a.m.	3 a.m.	4 a.m.	5 a.m.	6 a.m.	7 a.m.	8 a.m.	9 a.m.	10 a.m.	11 a.m.	12 a.m.
Mean equation												
<i>LWIND</i>	−0.0269***	−0.0357***	−0.0357***	−0.0284***	−0.0230***	−0.0320***	−0.0535***	−0.0609***	−0.0492***	−0.0408***	−0.0353***	−0.0357***
<i>LCONS</i>	0.3739***	0.4887***	0.4934***	0.3534***	0.3829***	0.4483***	0.5457***	0.6350***	0.5853***	0.4669***	0.4175***	0.4163***
<i>LINFLOW</i>	0.0272***			0.0342***	0.0323***	0.0252***	0.094*			0.0122**	0.0079	0.0265***
<i>LOUTFLOW</i>	0.0107***			0.1442***	0.0376***	0.0678***	−2.0411***	0.0947***	0.1100***	−1.002***	0.1009***	0.0922***
<i>ω</i>	0.3654	−0.7184*	0.0425***	0.0266***	−0.2207	−1.1344***	0.0072***	−2.6790***	−2.9935***	−1.3670***	−0.9355***	−1.3984***
<i>AR(1)</i>	0.9941***	0.9921***	0.9912***	0.9904***	0.9921***	0.9901***	0.9859***	0.9733***	0.9958***	0.9493***	0.9626***	0.9958***
<i>SAR(7)</i>		0.0213***		0.0287***	0.0261***	0.0542***	0.7307***	0.8367***	0.0422**	0.0512***	0.0457***	
<i>MA(1)</i>	0.1773***	−0.4658***	−0.5900***	−0.4201***	−0.4254***	−0.4952***	−0.5572***	−0.5536***	−0.7176***			−0.6699***
<i>SMA(1)</i>	−0.5865***		0.2058***				−0.6709***	−0.7586***	0.2748***			0.2756***
<i>SMA(7)</i>										−2.6469***	−2.7810***	−2.7240***
<i>D_{5_30_19}</i>												−1.4395***
<i>D_{5_12_19}</i>												−0.9870***
<i>D_{6_6_19}</i>												
<i>D_{2_10_20}</i>												
Variance equation												
<i>ω</i>	0.0196**	0.0156*	0.0207***	0.0250	0.0149***	0.0202***	0.0116	−0.0092	−0.0099	−0.0185	−0.0019	−0.0002
<i>α</i>	0.6669**	0.6349***	0.6823***	0.7487***	0.7351***	0.5789***	0.4805***	0.3248***	0.4289***	0.5658***	0.6151***	0.4849***
<i>β</i>	0.2844***	0.3290***	0.3281***	0.3059***	0.1970***	0.3534***	0.4879***	0.6688***	0.5760***	0.4676***	0.4453***	0.5147***
<i>LWIND</i>	0.0012***	0.0018***	−0.0031***	0.0006***	0.0003	−0.0006***	−0.0019***	−0.0012**	−0.0010***	−0.0004	−0.0003	−0.0004
<i>LCONS</i>	−0.0022**	−0.0028***	0.0015***	−0.0020***	−0.0007	0.0007	0.0049***	0.0023	0.0006	−0.0006	−0.0015	−0.001
<i>LINFLOW</i>	−0.0012***			−0.0011***	−0.0018***	−0.0021***	−0.0017**			0.0011**	0.0012	0.0006
<i>LOUTFLOW</i>	0.0005**	0.0002	0.0001	−2.23E-05	0.0007*	−0.0004	−0.0032***	−0.0003	0.0015*	0.0024***	0.0012**	0.0011
Diagnostic tests												
<i>R²</i>	0.6784	0.5905	0.5876	0.6428	0.6956	0.672	0.6424	0.6661	0.7121	0.6946	0.7406	0.8293
<i>AIC</i>	−1.5236	−1.2394	−1.2135	−1.4797	−1.8892	−1.6855	−1.0478	−0.9068	−1.1834	−1.164	−1.3245	−1.6355
<i>SIC</i>	−1.4693	−1.1951	−1.1694	−1.4286	−1.8381	−1.6344	−0.9933	−0.8591	−1.1323	−1.1129	−1.2733	−1.5745
<i>Log-likelihood</i>	1220.389	988.4174	972.2597	1179.503	1501.771	1341.455	840.5794	727.6474	946.3386	931.0287	1057.26	1310.901
Likelihood												
<i>Q_LB (15)</i>	7.617 [0.814]	11.816	10.857	14.668	30.331	40.128	31.151	33.141	32.894	66.993	44.082 [0.000]	31.176
		[0.461]	[0.541]	[0.260]	[0.002]	[0.000]	[0.001]	[0.000]	[0.001]	[0.001]	[0.002]	[0.002]
<i>Q_LB² (15)</i>	2.3901	3.3686	3.5241	1.5766	2.3377	3.846 [0.998]	5.7896	3.1403	4.4509	1.2475	0.953 [1.000]	15.087
	[1.000]	[0.999]	[0.999]	[1.000]	[1.000]	[0.983]	[0.983]	[1.000]	[0.996]	[1.000]	[0.445]	[0.445]
<i>ARCH(15)</i>	0.1655	0.2218	0.2312	0.1364	0.1577	0.2520	0.3762	0.2175	0.3060	0.0826	0.06254	1.0021
	[0.999]	[0.999]	[0.999]	[1.000]	[0.999]	[0.998]	[0.985]	[1.000]	[0.995]	[1.000]	[1.000]	[0.450]

Notes: Q_{LB} denotes the serial correlation test of standardized residuals; Q_{LB}^2 means the serial correlation test of the standardized squared of residuals; p -values are presented in “[]”, and ***, **, * stand for 1%, 5%, and 10% level of significance, respectively.

Table 2
Results of SARMAX/GARCH models between 1 p.m. and 12 p.m.

Variables	1 p.m.	2 p.m.	3 p.m.	4 p.m.	5 p.m.	6 p.m.	7 p.m.	8 p.m.	9 p.m.	10 p.m.	11 p.m.	12 p.m.
Mean equation												
<i>LWIND</i>	−0.0368***	−0.0359***	−0.0362***	−0.0314***	−0.0379***	−0.0392***	−0.0326***	−0.0261***	−0.0232***	−0.0186***	−0.0218***	−0.0295***
<i>LCONS</i>	0.4014***	0.4292***	0.4642***	0.4022***	0.4295***	0.4143***	0.3246***	0.3020***	0.2563***	0.2062***	0.1847***	0.3368***
<i>LINFLOW</i>	0.0404***	0.0313***	0.0133***	0.0308***	0.0098	0.0098	0.0334***	0.0535***	0.0391***	0.0574***	0.0524***	0.0216***
<i>LOUTFLOW</i>	0.0819***	0.0809***	0.0950***	0.0871***	0.1141***	0.1098***	0.0770***	0.0482***	0.0301***	0.0367***	0.0193***	0.0367***
ω	−1.1010***	−1.2651***	−1.4912***	−1.0576***	−1.2941***	−1.0966***	−0.2434	−0.0268	0.5050**	1.1218***	1.6803***	0.5454
<i>AR(1)</i>	0.9922***	0.9931***	0.9928***	0.9931***	0.9935***	0.9897***	0.9907***	0.9946***	0.9939***	0.9967***	0.9968***	0.9954
<i>SAR(7)</i>	0.5095***	0.5092***	0.4731***	0.0430***	0.0435***	0.5263***	0.5271***	0.0459**	0.7083***	0.9967***	0.9968***	0.9954
<i>SAR(30)</i>												
<i>MA(1)</i>	−0.4613***	−0.4544***	−0.4388***	−0.6246***	0.2386***	−0.4717***	−0.4297***			0.2163***	−0.0122**	−0.0132**
<i>SMA(1)</i>				0.2335***	−0.6494***			−0.5733***	−0.3467***	−0.5088***	−0.5011***	−0.5429***
<i>SMA(7)</i>	−0.4662***	−0.4683***	−0.4480***			−0.4872***	−0.4939***	0.2159***	−0.6900***		0.1575***	0.1433***
<i>D_{2,10,20}</i>										−1.6652***	−2.2400***	−2.5801***
<i>D_{2,9,20}</i>											−2.1726***	
<i>D_{4,21,17}</i>								−1.0670***	−1.6747***			
<i>D_{6,8,19}</i>									−1.3649***			
<i>D_{5,1,19}</i>												
<i>SD_{2,9,20}</i>												
<i>D_{5,30,19}</i>	−3.6815***	−2.9978***	−2.2387***		−0.5810***							
Variance equation												
ω	0.012	0.0095	0.0147	0.0098	0.0065	0.009	0.0146***	0.0122***	0.0079***	0.0061	0.0078	0.0245***
α	0.6089***	0.6001***	0.6591***	0.5123***	0.4866***	0.4459***	0.5675***	0.4332***	0.3763***	0.4807***	0.4774***	0.6275***
β	0.3636***	0.3551***	0.3320***	0.4656***	0.5352***	0.5366***	0.4823***	0.5737***	0.6063***	0.4595***	0.3517***	0.2797***
<i>LWIND</i>	−0.0009***	−0.0006*	−0.0003	0.0001	−0.0001	−0.0005	−0.0005***	−0.0002	−4.85E-05	0.0003***	0.0006***	0.0009***
<i>LCONS</i>	−0.0012	−0.0008	−0.0011	−0.0008	−0.0007	−0.0002	−0.0006	−0.0006	−0.0004	−0.0005	−0.0009	−0.0025***
<i>LINFLOW</i>	−9.98E-05	−0.0003	−0.0001	0.0005	0.0013***	0.0018***	0.0013***	0.0004	−9.34E-05	−0.0002	−0.0009***	−0.0009***
<i>LOUTFLOW</i>	0.001	0.0008	7.23E-05	−0.0008	−0.0011	−0.0021**	−0.0018***	−0.0010***	−0.0004	−0.0001	0.0006**	0.0003
Diagnostic tests												
R^2	0.8156	0.8295	0.8123	0.8113	0.7918	0.7554	0.7364	0.7733	0.8396	0.8575	0.76	0.6891
AIC	−1.6339	−1.7491	−1.7376	−1.6626	−1.4088	−1.4628	−1.8261	−2.3574	−2.6413	−2.6833	−2.2618	−1.8439
SIC	−1.576	−1.6912	−1.6797	−1.6047	−1.3509	−1.4083	−1.7716	−2.2995	−2.5800	−2.629	−2.1997	−1.7853
Log-likelihood	1302.903	1393.557	1384.495	1325.492	1125.715	1167.198	1453.16	1872.274	2096.662	2137.183	1772.021	1446.971
$Q_{LB}(15)$	21.037	25.66 [0.007]	24.814	23.091	20.365	26.863	19.927	5.3329	13.613 [0.255]	16.012	15.202	7.1199
$Q_{LB}^2(15)$	[0.033]	[0.010]	[0.010]	[0.017]	[0.041]	[0.005]	[0.046]	[0.914]	[0.005]	[0.191]	[0.173]	[0.789]
$Q_{LB}^3(15)$	0.7165	9.9138 [0.825]	19.497	1.6346	2.9356	3.3074	4.2992	0.0997	1.0416 [1.000]	5.5117	3.6021	1.1649
ARCH(15)	[1.000]	[0.192]	[1.000]	[1.000]	[1.000]	[0.999]	[0.997]	[1.000]	[0.999]	[0.987]	[0.999]	[1.000]
	0.0464	0.6547	1.3037	0.1070	0.1897	0.2184	0.2829	0.0665	0.06751	0.3671	0.2326	0.0833
	[1.000]	[0.8303]	[0.191]	[1.000]	[0.999]	[0.999]	[0.997]	[1.000]	[1.000]	[0.987]	[0.999]	[1.000]

Notes: Q_{LB} denotes the serial correlation test of standardized residuals; Q_{LB}^2 means the serial correlation test of the standardized squared of residuals; the p-values are presented in “[]”; and ***, **, * stand for 1%, 5% and 10% level of significance, respectively.

ARCH-LM tests (displayed in Tables 1 and 2). The results suggested the rejection of serial autocorrelation in the standardized squared residuals up to 15 lags, for all hours. This finding is important, as it indicates that the volatility is being properly explained. Please note that the hypothesis of autocorrelation in squared residuals was rejected beyond 15 lags in the majority of estimations. However, the maximum number of lags that guaranteed the rejection of autocorrelation in all hours was 15. Nonetheless, the choice of 15 lags was justified, as it is reasonable to expect that the electricity system would return to equilibrium after 15 days, and the existence of autocorrelation beyond 15 lags should, therefore, not be a concern. Moreover, the results also indicated the non-existence of ARCH effects in residuals up to 15 lags, for all hours.

With regard to the autocorrelation of standardized residuals, the outcomes suggested the presence of autocorrelation in models in the hours starting from 5 a.m. to 7 p.m. This autocorrelation issue should only be a concern if there is a market-splitting mechanism, which might incentivize investors to benefit from arbitrage opportunities. The autocorrelation in standardized residuals could actually suggest potential inefficiencies in the daily forecasts for the day-ahead electricity market and/or in the management of interconnecting lines (for instance, if they become congested, market-splitting will probably occur). Another diagnostic test that ensured the accuracy of the results was the theoretical Quantile-Quantile plots.³ Most of the data in these plots lay on a theoretical line, which confirmed that the residuals followed a Student-*t* distribution. Finally, the overall stationarity of models was confirmed, because the condition $\alpha + \beta < 1$ was verified and, both the inverted AR and MA roots were stationary. However, there were some exceptions, i. e., hours when the aforementioned condition proved not to be true. In these cases, the volatility in electricity prices was likely to reflect a permanent impact, which may indicate that unconditional volatility does not die away. These hours coincided precisely with periods when the electricity market either seemed to be highly dynamic (at 5 p.m.; and in periods between 9 a.m. and 11 a.m.; and 7 p.m. and 8 p.m.), or when the dynamics were relatively low, (3 a.m. and 4 a.m.).

The results of the estimation show that electricity produced from wind power reduces the day-ahead electricity price, underlining the MOE for all of the 24-h of the day. The magnitude of the impact of electricity production from wind power seemed homogeneous among all estimations. Nonetheless, a slightly increase in the impact of wind power on electricity prices was observed between 7 a.m. and 9 a.m. The highest impact was observed at 8 a.m., in other words, a 1% increase in electricity produced from wind power decreased the day-ahead electricity price by around 0.0609% (Table 1). Meanwhile, the volatility of the day-ahead electricity price was only increased by electricity produced from wind power during a few hours of the day (six occurrences were found). Therefore, there seems to be a higher probability of wind power decreasing volatility in electricity prices than increasing it. However, no clear pattern of volatility was identified, so that, at off-peak consumption hours, for example, wind power production appeared to both increase and decrease the volatility of electricity prices.

According to the results, electricity consumption had a prominent impact on the mean of the day-ahead electricity price, increasing it in all hours. As was noted in the case of wind power, the strongest impact of electricity consumption also occurred in the ramp-up to the peak period of consumption. A 1% rise in electricity consumption at 8 a.m., augmented the day-ahead electricity price by around 0.6350% (Table 1). On the other hand, electricity consumption did not appear to be statistically significant in most of the variance equation models,

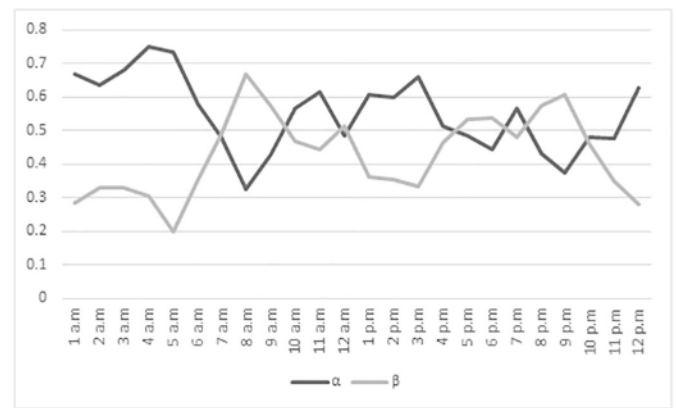


Fig. 5. Results of the GARCH error parameter and GARCH lag parameter for each of the 24 estimations.

which may suggest it has no direct and substantial impact on the volatility of electricity prices.

Boosting the inflow of electricity appeared to increase the day-ahead price of electricity in most hours of the day. The largest impact occurred at 10 p.m., when an increase of 1% in electricity inflow augmented the day-ahead electricity price around 0.0574% (Table 2). The magnitude of electricity inflow was, actually, roughly similar for all hours. The outflow of electricity also positively influenced the day-ahead electricity price. Generally, its impact was greater than that of electricity inflow, and oscillated freely throughout the 24 h of the day. The greatest impact of electricity outflow on the day-ahead electricity price was found to be at 4 a.m., by increasing it 0.1442% (Table 1).

Considering the impact of electricity inflow and outflow on the volatility of electricity prices, both variables seems to have positive and negative effects. Generally speaking, the same effect was not found simultaneously for both variables. For instance, when electricity inflow was increasing volatility, outflow appeared to be decreasing it, and vice-versa. Briefly, compared to the results obtained from wind power and electricity consumption, the impact of both the inflow and outflow of electricity on volatility was fairly negligible. In fact, there was evidence these variables were not statistically significant in a substantial number of estimations, mostly at peak hours. However, it is worth mentioning the positive statistically-significant effects of electricity inflow in variance equations, between 5 p.m. and 7 p.m., although. It should be noted that this variable was not significant from 9 p.m. to 5 p.m. This effect could be explained by an increasing need to satisfy consumption peaks.

For all hours, the volatility of day-ahead electricity price displayed a substantial degree of persistence, i.e., $\alpha_1 + \beta_1$ was nearly 1. These results are consistent with the notion that the volatility of day-ahead electricity prices have long memory, rather than the exponential decay of a shock to an average level implied by short memory. The largest level of persistence took place at 12 a.m. ($\alpha_1 + \beta_1 = 0.9936$) (Table 1), while the lowest level was found at 11 p.m. ($\alpha_1 + \beta_1 = 0.8291$) (Table 2). Therefore, it is reasonable to conclude that the behaviour of future volatility at 12 a.m. was highly reliant on past volatility, and this was also confirmed by the outcome $\beta_1 = 0.6688$. Meanwhile, due to the low persistence level noted at 11 p.m., a low dependence of future volatility on past shocks was also expected, and confirmed by the result $\beta_1 =$

³ Once again, to save space, the Quantile-Quantile plots are not presented, but are available upon request.

0.3517. After highlighting the hours when volatility had a greater or lower persistence of volatility, and thus long or short memory respectively, an overview of the α_1 and β_1 coefficients was justified. Fig. 5 summarizes the values for α_1 and β_1 obtained in the estimations for each of the 24 h.

From Fig. 5 it is clear that the values of α_1 and β_1 did not move in the same direction throughout the day. This means that when the GARCH error parameter increased, the GARCH lag parameter appeared to reduce, and vice-versa. This relationship clearly demonstrates the cyclical nature of the dynamics of the electricity system in the SE3 bidding zone. In most estimations, the α_1 parameter had greater explanatory power for the volatility, than the β_1 . However, at peak hours, when high prices are likely to persist, mainly because of increased electricity consumption, the transmission of volatility (measured by β_1) seemed to be higher. Moreover, the impact on volatility of new shocks (assessed by α_1) seemed less marked in peak than off-peak periods. Metaphorically speaking, it is like comparing the impact of ripples on a lake to their much smaller impact on an ocean.

6.2. Discussion

Wind and solar PV generation have been increasing exponentially, and electricity systems have struggled to deal with the complexities and uncertainties due to their highly variable production of electricity. Interconnected capacity has been introduced to address local imbalances between supply and demand, but the overall flexibility of electrical power systems is undoubtedly a key factor for ensuring their reliability. Due to the low capacity of flexible generation in the SE3 BZ, the flexibility of this electricity system is dependent on its intensified cross-border electricity interconnections. The cross-border flow of electricity is a major mechanism being used to guarantee electricity during shocks in supply or demand. However, the impact of such networks on the efficient management of fluctuations in electricity systems has been insufficiently studied. With this in mind, this research examined the SE3 BZ to analyse the impact of the inflow and outflow of electricity on the volatility of the day-ahead electricity price. Furthermore, this study studied the MOE of wind power on a system with a substantial proportion of its electricity generated by nuclear power. The results show that electricity production from wind power, and the inflow and outflow of electricity, all impact both the mean and variance of day-ahead electricity prices in the SE3 BZ.

This study shows the existence of a MOE from wind power in the SE3 BZ. The magnitude of this effect was found to be stable throughout the 24 h of the day. This finding may be consistent with the fact that a large part of the electricity in the SE3 BZ is nuclear generated. Like wind energy, nuclear power provides baseload power with low operational costs, but large capital costs. Given this, it was reasonable to expect the MOE from wind power would be eclipsed by that from nuclear power, which plays such a prominent role in the SE3 BZ generation mix. In other words, the magnitude of the MOE from wind power might be far more higher in a different electricity generation mix consisting of baseload and backup energy sources with high marginal costs. Janda (2018) also justified the relatively low MOE of solar PV power in Slovakia, by the predominant influence of nuclear power there. From this point of view,

nuclear electricity sources appear to be a promising source for efficiently and expeditiously accommodating intermittent RES, by hedging against their disruptive impact on the stability of electricity systems (at certain hours of the day). However, nuclear energy provides baseload power plants with a rigid production portfolio, which may decrease the flexibility of electricity systems. The ongoing decommissioning of nuclear power plants in the SE3 BZ, may be increasing the tendency to import electricity from neighbouring market areas and, thus, lead to rises in domestic electricity prices. This phenomenon was confirmed in the French electricity system by Rinne (2019), who found that a downward shock in nuclear capacity increased the price of electricity.

The results indicate the positive impact of both electricity inflows and outflows on day-ahead electricity prices, in the majority of the 24 estimations. Electricity imports and exports can trigger two possible scenarios in the electricity market, depending on whether or not electricity interconnections are congested. In the first scenario, when interconnecting power lines become congested, electricity exports causes price differences between neighbouring markets, and may lead to market splitting phenomena (Ringler et al., 2017). This could occur when there is a simultaneous increase in electricity demand in the SE3 BZ, and high supply of electricity from RES in other BZs. Accordingly, electricity is likely to flow from zones where its price is low (exporting area) to zones where its price is high (importing area). Consequently, the inflow of electricity will increase the electricity price in the importing country. Conversely, in the second scenario, when the interconnecting power lines are uncongested, electricity is exported to the neighbouring market areas until electricity prices become roughly equalized (Keles et al., 2020). In contrast to the first scenario, this adjustment in electricity prices will probably increase prices in the exporting zone, while decreasing them in the importing zone.

Briefly, according to the data published on the Nord Pool Official website, the biggest recipient of electricity exports from the SE3 BZ, is the FI BZ. The FI BZ generally has the highest day-ahead electricity price of all the Nordic Baltic countries, and transmission congestion between FI and SE3 is rare. Therefore, the positive impact of electricity outflow on the mean of the day-ahead electricity price (observed in most of the 24 h) is not surprising. Looking at electricity imports into the SE3 BZ, a significant part of them come from the SE2 and NO1 BZs. Electricity interconnections from these bidding zones are often congested, increasing the likelihood of market splitting. Thus, it was reasonable to expect an increase in day-ahead electricity price due to electricity imports, and this was confirmed in the results.

The inflow and outflow of electricity appears to have a different effect on volatility than on the mean of day-ahead electricity prices. Neither inflow nor outflow appear to influence price volatility, as shown by the lack of statistical significance of their coefficients in the variance equations in most hours, particularly during peak periods. This is more likely to be a reflection of the high dependence of the SE3 BZ on electricity imports to satisfy consumption peaks, than of any strategy to manage fluctuations in the system. In contrast, residual volatility may have been neutralized by the decoupling of the bidding areas. A similar conclusion was reached by Sapio and Spagnolo (2020) to explain the Italian context. Meanwhile, the negligible impact of electricity outflow on the volatility of electricity prices is consistent with the finding that

electricity exports from the SE3 BZ are more likely to occur during off-peak hours. Considering the impact of wind power on the day-ahead electricity price, the results show wind affects price volatility in both directions, i.e., positively and negatively.

Nuclear power is a good source of reasonably-priced baseload electricity, but is less suited to providing flexible backup capacity for intermittent sources, or to satisfy peaks of consumption. Thus, electricity systems with a high share of nuclear production may block, or at least constrain, the introduction of alternative energy sources that require this flexibility. However, if such nuclear-dominated systems also have a high capacity of interconnections, this problem may be mitigated, and RES producers will be more likely to invest in additional capacity, in the medium- and long-term. This study of the SE3 BZ reveals a system where the day-ahead electricity price seems to increase with the flow between zones, regardless of whether the electricity is imported or exported. These high prices may incentivize investment in RES, as opportunities to maximize profits are greater in an electricity system with higher electricity prices. However, the flexibility of the SE3 BZ system was shown to be low, as it is highly reliant on electricity produced in the interconnected areas. This actually makes the electrical system in this BZ less predictable and, thus, highly susceptible to new shocks, as suggested by the results.

In summary, the mean of the day-ahead electricity price in the SE3 BZ seems to be influenced by wind power production, and by the inflow and outflow of electricity. Conversely, the volatility of the electricity price is hardly influenced at all by the cross-border flow of electricity. Meanwhile, wind power influences this volatility both positively and negatively. There is an urgent need to speed up the development of RES in the SE3 BZ, to replace nuclear generation, and to incentivize the installation of backup capacity. Otherwise, in the future, unexpected sharp increases in electricity consumption in the SE3 BZ could potentially lead to hours of electricity scarcity, reduced security of supply, or even vertiginous increases in electricity prices.

7. Conclusion

This paper undertook an analysis of the impact of wind power, and of electricity inflow and outflow, on both the mean and volatility of the day-ahead electricity price in the SE3 BZ. Each hour of the day was analysed individually (i.e., by estimating up to 24 different models) during the period of 1 January 2016 to 30 April 2020, to take into account strong potential seasonal effects. The main outcomes of this study suggest a permanent MOE from wind power during all 24 h of the day, and the positive impact of both the import and export of electricity on the mean of day-ahead electricity price. Besides this, unprecedented and/or new shocks in the electricity price time-series were found to have higher explanatory power than past volatility, for the time-varying volatility of electricity prices in the SE3 BZ. The high dependence of the system on electricity imports and exports could explain this finding. Furthermore, the volatility of electricity prices seems to be highly

persistent at all hours, which indicates that time-varying volatility is unlikely to die away.

This analysis proves that market integration and cross-border electricity transmission play a key role in explaining the formation of domestic electricity prices. Optimal scheduled strategies to deal with unexpected events need to be further developed to avoid the worst effects of unprecedented shocks on the volatility of day-ahead electricity price and thus, improve the stability of the system as a whole. One energy source with high potential to provide back up and mitigate the variability of RES, is hydro power, and this source could also help improve the flexibility of the electricity system in the SE3 BZ. However, unless massive additional capacity of interconnections is installed, the market-splitting tendency will persist, and trading strategies will wield their high potential for balancing the system.

In general, electricity systems with a high proportion of nuclear generation are likely to have higher electricity prices, and this encourages strategies to transition towards becoming green and sustainable electricity systems. However, these strategies are only feasible in an integrated electricity market with high interconnectivity. In systems with substantial electricity autarky (and thus, weak interconnecting capacity), such as the French electricity system, nuclear power may constraint the introduction of other energy sources, specially of wind and solar PV power. Firstly, because electricity prices are likely to be low, and secondly, because nuclear power decreases the flexibility of electricity systems and, accordingly, instantaneous backup capacity will also be rather low. This will lessen the willingness of producers to invest in additional RES capacity.

With a large part of its electricity generated from nuclear power and strong interconnectivity, the SE3 BZ, in terms of its internal system goals, could represent an efficient strategy for incorporating more wind power (as revealed by the small current MOE of wind). However, taking a broader view, the findings of this research highlight the high vulnerability of the SE3 BZ system to shocks, which may increase uncertainty in the overall electricity market. Furthermore, the low backup capacity in the SE3 BZ may reduce the system's predictability, as it is heavily dependent on electricity imports. Consequently, unless RES producers are happy to assume this risk, they may be unwilling to increase their investment in RES capacity.

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Appendix

Table A.1
Descriptive statistics.

	A.M.												
		1 a.m.	2 a.m.	3 a.m.	4 a.m.	5 a.m.	6 a.m.	7 a.m.	8 a.m.	9 a.m.	10 a.m.	11 a.m.	12 a.m.
Mean	LPRICE	3.2959	3.2631	3.2493	3.2690	3.3539	3.4546	3.5744	3.6369	3.6239	3.5706	3.5470	3.5179
	LWIND	6.3752	6.3685	6.3610	6.3530	6.3338	6.2975	6.2531	6.1993	6.1621	6.1354	6.1329	6.1478
	LCONS	9.0464	9.0354	9.0329	9.0379	9.0671	9.1424	9.2171	9.2427	9.2591	9.2724	9.2753	9.2719
	LINFLOW	7.1327	7.1021	7.0966	7.1578	7.4820	8.0578	8.3389	8.4708	8.5252	8.5350	8.5265	8.4896
	LOUTFLOW	7.9765	7.9621	7.9532	7.9822	8.0929	8.3077	8.3997	8.4277	8.4277	8.4042	8.3948	8.3721
Max.	LPRICE	4.0513	4.0422	4.0332	4.0395	4.1117	4.3435	5.3031	5.4533	5.3279	5.2608	5.1232	5.0169
	LWIND	7.7895	7.8026	7.8022	7.8026	7.8083	7.8055	7.7973	7.7782	7.7740	7.7811	7.7985	7.8038
	LCONS	9.5324	9.5262	9.5556	9.5677	9.6073	9.6844	9.7514	9.8101	9.7929	9.7751	9.7473	9.7298
	LINFLOW	8.9219	8.9111	8.9134	8.9364	9.0073	9.0991	9.1412	9.1277	9.1439	9.1423	9.1321	9.1270
	LOUTFLOW	8.8841	8.8878	8.8878	8.8841	8.9531	8.9396	8.9862	8.9534	8.9331	8.9274	8.9252	8.9367
Min.	LPRICE	-0.2231	-0.1054	-0.0726	0.0583	0.6206	0.9821	1.0403	1.0750	1.1378	0.7930	0.7467	0.7080
	LWIND	3.5264	3.5264	3.6109	3.6376	3.7377	3.2958	2.9957	2.7726	2.7726	2.7081	2.9444	3.0910
	LCONS	8.6440	8.6223	8.6103	8.5999	8.5824	8.5901	8.6214	8.6728	8.7349	8.7932	8.8225	8.8311
	LINFLOW	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	5.1556	5.2983	5.2983
	LOUTFLOW	0.0000	0.0000	0.0000	0.0000	5.0720	5.4493	6.0398	6.8516	6.8059	6.9634	7.0797	7.0157
Std. Dev.	LPRICE	0.5088	0.5348	0.5506	0.5436	0.4818	0.4396	0.4183	0.4073	0.3966	0.4186	0.4271	0.4362
	LWIND	0.7632	0.7637	0.7662	0.7708	0.7864	0.8193	0.8636	0.9074	0.9427	0.9688	0.9699	0.9582
	LCONS	0.1985	0.2033	0.2080	0.2167	0.2271	0.2384	0.2400	0.2261	0.2083	0.1966	0.1895	0.1854
	LINFLOW	1.4920	1.4188	1.4044	1.4330	1.3086	1.0244	0.9149	0.7148	0.5508	0.4708	0.4614	0.4736
	LOUTFLOW	0.5427	0.5481	0.5729	0.5373	0.4134	0.3808	0.3566	0.3268	0.3149	0.3095	0.3087	0.3140
Skewness	LPRICE	-1.9932	-1.9976	-2.0153	-2.0197	-1.9053	-1.8597	-1.4074	-1.0327	-1.1680	-1.6621	-1.8271	-1.8765
	LWIND	-0.5049	-0.5104	-0.5209	-0.5263	-0.5430	-0.5732	-0.6356	-0.6728	-0.6808	-0.6756	-0.6029	-0.6029
	LCONS	-0.0030	-0.0246	-0.0477	-0.0936	-0.1538	-0.2320	-0.2969	-0.2706	-0.2092	-0.1514	-0.1085	-0.0875
	LINFLOW	-3.0686	-3.0871	-3.0802	-3.2255	-3.6303	-3.9757	-4.5921	-5.1893	-4.1406	-2.1353	-1.9806	-1.8714
	LOUTFLOW	-4.8782	-4.9395	-5.6697	-5.1361	-0.8097	-1.1187	-1.3807	-1.2636	-1.2852	-1.0999	-0.9969	-0.8884
Kurtosis	LPRICE	8.6376	8.4401	8.4281	8.3025	7.8894	8.1581	8.1702	7.5838	8.0704	9.7094	10.0345	9.7676
	LWIND	2.9440	2.9882	3.0151	3.0027	3.0021	3.0062	3.0844	3.1202	3.0767	3.0164	2.9169	2.8338
	LCONS	1.8527	1.8506	1.8434	1.8458	1.8912	2.0608	2.2496	2.3030	2.2715	2.2159	2.1707	2.1466
	LINFLOW	14.6421	15.2695	15.4704	16.0773	20.3400	28.3575	37.2485	53.0650	44.4382	10.2959	9.2370	8.6330
	LOUTFLOW	64.8947	63.9356	74.1424	68.8420	5.4834	5.7849	5.8646	4.7499	5.0462	4.3866	3.9409	3.8074
Jarque-Bera	LPRICE	3142.5200	3002.9250	3013.1080	2928.8790	2532.9330	2665.6830	2284.2640	1666.1760	2054.3240	3695.7460	4142.1040	3947.3900
	LWIND	67.4147	68.6910	71.5598	73.0369	77.7278	86.6446	106.9974	120.2902	122.5903	120.3530	108.3775	97.6743
	LCONS	86.7721	87.2507	88.7701	90.1280	87.2666	72.3406	60.3621	51.3296	46.5311	46.5697	48.4383	50.0190
	LINFLOW	11,447.1700	12,405.8900	12,752.2200	14,015.9300	23,294.4000	46,551.9900	82,877.5000	172,320.5000	117,707.8000	47,10.8840	3598.4460	3014.9820
	LOUTFLOW	258,797.9000	251,191.3000	342,095.4000	292,715.2000	579.4082	841.2153	1043.5120	622.8697	711.5124	445.7242	320.4002	251.0594
Obs.	LPRICE	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582
	LWIND	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582
	LCONS	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582
	LINFLOW	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582
	LOUTFLOW	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582

P.M.

(continued on next page)

Table A.1 (continued)

	P.M.											
	1 p.m.	2 p.m.	3 p.m.	4 p.m.	5 p.m.	6 p.m.	7 p.m.	8 p.m.	9 p.m.	10 p.m.	11 p.m.	12 p.m.
Mean	LPRICE 3.4910	3.5093	3.5107	3.4948	3.5523	3.5717	3.5515	3.5013	3.4670	3.4537	3.3738	3.3376
	LWIND 6.1732	6.2009	6.2231	6.2463	6.2656	6.2816	6.2971	6.3140	6.3390	6.3688	6.3854	6.3772
	LCONS 9.2647	9.2577	9.2578	9.2608	9.2728	9.2719	9.2604	9.2401	9.2081	9.1640	9.1146	9.0710
	LINFLOW 8.4485	8.4119	8.4137	8.4577	8.5508	8.6058	8.5909	8.5001	8.3733	8.1756	7.8243	7.3155
	LOUTFLOW 8.3570	8.3430	8.3490	8.3766	8.4463	8.4993	8.5000	8.3816	8.2968	8.1510	8.0161	8.0161
Max.	LPRICE 5.0757	5.0815	4.6471	5.2982	5.2988	5.2977	4.6046	4.4129	4.2872	4.1890	4.0678	4.0704
	LWIND 7.8026	7.8022	7.8026	7.8314	7.8071	7.8116	7.8136	7.8104	7.8030	7.7998	7.7928	7.7828
	LCONS 9.7064	9.7033	9.7135	9.7314	9.7451	9.7419	9.7350	9.7097	9.6795	9.6363	9.5922	9.5531
	LINFLOW 9.1367	9.1395	9.1395	9.1286	9.1499	9.1662	9.1526	9.1338	9.0931	9.0496	8.9954	8.9346
	LOUTFLOW 8.9449	8.9411	8.9242	8.9942	9.0325	9.0047	9.0261	9.0152	9.0248	9.0188	8.8841	8.8704
Min.	LPRICE -0.2357	0.4574	0.9002	0.5878	0.8544	1.1019	1.3584	1.4085	1.1848	0.8329	0.0392	-0.0943
	LWIND 3.2581	3.3322	3.3673	3.3673	3.4657	3.5264	3.6109	3.6889	3.7612	3.8286	3.8067	3.5264
	LCONS 8.8305	8.8248	8.8177	8.8164	8.8226	8.8191	8.8045	8.7843	8.7664	8.7655	8.7315	8.6839
	LINFLOW 5.2983	0.0000	0.0000	5.3836	5.4651	6.0191	6.1527	5.0106	4.4659	0.0000	0.0000	0.0000
	LOUTFLOW 7.0868	7.0500	6.9863	7.0756	7.1874	7.1498	6.9474	6.5013	6.1088	5.6747	5.4939	0.0000
Std. Dev	LPRICE 0.4629	0.4444	0.4486	0.4776	0.4582	0.4253	0.3973	0.3991	0.4100	0.4149	0.4642	0.4984
	LWIND 0.9396	0.9203	0.9056	0.8890	0.8732	0.8572	0.8379	0.8178	0.7922	0.7617	0.7432	0.7664
	LCONS 0.1842	0.1870	0.1955	0.2054	0.2121	0.2112	0.2068	0.1998	0.1920	0.1851	0.1851	0.1912
	LINFLOW 0.5023	0.5696	0.6048	0.5107	0.4537	0.3884	0.3562	0.3852	0.4483	0.5816	0.8381	1.4112
	LOUTFLOW 0.3263	0.3366	0.3396	0.3296	0.3018	0.2734	0.2638	0.2911	0.3346	0.3880	0.4301	0.5005
Skewness	LPRICE -2.1526	-1.9398	-1.8605	-1.8468	-1.5625	-1.6760	-1.8093	-1.9486	-2.0374	-2.0130	-2.0836	-2.0869
	LWIND -0.5701	-0.5475	-0.5293	-0.5039	-0.4904	-0.4813	-0.4473	-0.4253	-0.4209	-0.4066	-0.4098	-0.4934
	LCONS -0.0718	-0.0618	-0.0496	-0.0233	-0.0319	-0.0655	-0.1101	-0.0655	-0.1221	-0.0430	0.0126	0.0164
	LINFLOW -1.7854	-3.4739	-4.4042	-1.5326	-1.6632	-1.7293	-1.6327	-2.0624	-2.1448	-3.4722	-3.6992	-3.4410
	LOUTFLOW -0.8109	-0.7583	-0.7817	-0.7596	-0.9620	-1.1777	-1.3015	-1.3645	-1.3816	-1.3650	-1.0777	-3.5687
Kurtosis	LPRICE 11.6197	9.6901	8.9117	9.6993	9.1881	9.5470	9.4897	9.4734	9.5732	9.1078	9.2740	8.9979
	LWIND 2.7586	2.7073	2.6450	2.5678	2.5425	2.5486	2.4933	2.4908	2.5666	2.6336	2.7169	2.8704
	LCONS 2.1390	2.1144	2.0570	1.9617	1.8958	1.8737	1.9164	1.9898	2.0137	1.9818	1.9372	1.8715
	LINFLOW 7.9181	36.5151	51.0045	6.1969	7.0208	7.7162	7.3904	12.1051	12.1184	33.4927	29.0580	17.6012
	LOUTFLOW 3.5465	3.4132	3.4758	3.2199	3.7437	4.5927	5.3217	5.9241	6.1685	6.0254	5.1001	47.7892
Jarque-Bera	LPRICE 6119.2830	3942.3100	3216.3400	3857.6920	3167.8090	3566.0080	3639.3760	3763.4500	3942.5810	3527.5230	3739.3060	3519.5610
	LWIND 89.5459	84.6963	82.1827	79.2500	77.2075	74.5010	69.6694	64.7878	59.0847	52.4421	49.5664	65.2969
	LCONS 50.2237	52.7061	59.2618	71.2093	80.6364	84.7511	80.5971	72.8026	68.0512	68.8305	74.5017	84.0205
	LINFLOW 2434.8050	77.223.6100	157.014.6000	1293.0080	1795.0490	2254.6280	1973.4290	6586.1860	6693.6330	64.468.4600	48.366.5900	17.174.9900
	LOUTFLOW 193.0684	162.8720	176.0282	155.3380	280.4713	532.9093	801.9278	1054.5510	1165.0510	1094.5970	596.9823	135.591.5000
Obs.	LPRICE 1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582
	LWIND 1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582
	LCONS 1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582
	LINFLOW 1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582
	LOUTFLOW 1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582	1582

Notes: "Max." And "Min" signifies maximum and minimum, accordingly, "Std. Dev" means standard deviation, "Obs" denotes the number of observations included in the analysis and the *p*-values are shown in "[]".

Table A.2
Robinson estimates of fractional differencing parameter.

	Variables	Est. d.	Std. Err.	$t(H_0 : d = 0)$	$P > t $
1 a.m.	LPRICE	0.4516	0.0235	19.2506	0.000
	LWIND	0.2767	0.0235	11.7653	0.000
	LCONS	0.7709	0.0248	31.0322	0.000
	LINFLOW	0.3637	0.0239	15.2046	0.000
	LOUTFLOW	0.3890	0.0236	16.5013	0.000
2 a.m.	LPRICE	0.4281	0.02409	17.7724	0.000
	LWIND	0.2772	0.0239	11.6127	0.000
	LCONS	0.7646	0.0240	31.8032	0.000
	LINFLOW	0.3445	0.0241	14.3033	0.000
	LOUTFLOW	0.3623	0.0234	15.4559	0.000
3 a.m.	LPRICE	0.4372	0.0252	17.3176	0.000
	LWIND	0.2771	0.0239	11.5942	0.000
	LCONS	0.7539	0.0249	30.2182	0.000
	LINFLOW	0.3433	0.0221	15.5507	0.000
	LOUTFLOW	0.3250	0.0233	13.9651	0.000
4 a.m.	LPRICE	0.4504	0.0237	19.0254	0.000
	LWIND	0.2775	0.0230	12.0569	0.000
	LCONS	0.7448	0.0240	31.0503	0.000
	LINFLOW	0.3695	0.0219	16.8399	0.000
	LOUTFLOW	0.3337	0.0237	14.0774	0.000
5 a.m.	LPRICE	0.4416	0.0249	17.7671	0.000
	LWIND	0.2901	0.0235	12.3702	0.000
	LCONS	0.7295	0.0236	30.9041	0.000
	LINFLOW	0.3407	0.0238	14.3074	0.000
	LOUTFLOW	0.4165	0.0245	17.0216	0.000
6 a.m.	LPRICE	0.3409	0.0253	13.4692	0.000
	LWIND	0.3073	0.0234	13.1558	0.000
	LCONS	0.6815	0.0251	11.6739	0.000
	LINFLOW	0.2925	0.0251	11.6739	0.000
	LOUTFLOW	0.3766	0.0256	14.6961	0.000
7 a.m.	LPRICE	0.3637	0.0244	14.9327	0.000
	LWIND	0.3274	0.0246	13.2941	0.000
	LCONS	0.6133	0.0248	24.6848	0.000
	LINFLOW	0.2155	0.0250	8.6359	0.000
	LOUTFLOW	0.3652	0.0249	14.6707	0.000
8 a.m.	LPRICE	0.3973	0.0250	15.8999	0.000
	LWIND	0.3340	0.0259	12.9129	0.000
	LCONS	0.6057	0.0248	24.3892	0.000
	LINFLOW	0.2289	0.0261	8.7699	0.000
	LOUTFLOW	0.3594	0.0253	14.2126	0.000
9 a.m.	LPRICE	0.4340	0.0239	18.1347	0.000
	LWIND	0.3330	0.0261	12.7768	0.000
	LCONS	0.6473	0.0256	25.3336	0.000
	LINFLOW	0.2791	0.0249	11.2220	0.000
	LOUTFLOW	0.3481	0.0239	14.5578	0.000
10 a.m.	LPRICE	0.4436	0.0238	18.6352	0.000
	LWIND	0.3304	0.0251	13.1593	0.000
	LCONS	0.6714	0.0262	25.6759	0.000
	LINFLOW	0.3660	0.0239	15.2919	0.000
	LOUTFLOW	0.3955	0.0230	17.2135	0.000
11 a.m.	LPRICE	0.4436	0.0252	17.5834	0.000
	LWIND	0.3310	0.0246	13.4389	0.000
	LCONS	0.6940	0.0258	26.9464	0.000
	LINFLOW	0.3889	0.0250	15.5798	0.000
	LOUTFLOW	0.3740	0.0227	16.4647	0.000
12 a.m.	LPRICE	0.4292	0.0243	17.6441	0.000
	LWIND	0.3256	0.0247	13.2082	0.000
	LCONS	0.6932	0.0259	26.8143	0.000
	LINFLOW	0.4247	0.0257	16.5230	0.000
	LOUTFLOW	0.3928	0.0246	15.9764	0.000
1 p.m.	LPRICE	0.4054	0.0238	17.0329	0.000
	LWIND	0.3168	0.0237	13.3462	0.000
	LCONS	0.7004	0.0258	27.1300	0.000
	LINFLOW	0.4287	0.0243	17.6231	0.000
	LOUTFLOW	0.3837	0.0247	15.5400	0.000
2 p.m.	LPRICE	0.4328	0.0238	18.1844	0.000
	LWIND	0.3162	0.0238	13.2737	0.000
	LCONS	0.7058	0.0254	27.8288	0.000
	LINFLOW	0.3726	0.0238	15.6712	0.000
	LOUTFLOW	0.3973	0.0244	16.2760	0.000
3 p.m.	LPRICE	0.4414	0.0241	18.3177	0.000
	LWIND	0.3196	0.0240	13.3130	0.000
	LCONS	0.7243	0.0261	27.7277	0.000
	LINFLOW	0.3294	0.0234	14.0548	0.000

(continued on next page)

Table A.2 (continued)

	Variables	Est. d.	Std. Err.	$t(H_0 : d = 0)$	$P > t $
4 p.m.	LOUTFLOW	0.4118	0.0252	16.3485	0.000
	LPRICE	0.4565	0.0234	19.5017	0.000
	LWIND	0.3204	0.0235	13.6137	0.000
	LCONS	0.7542	0.0260	28.9583	0.000
	LINFLOW	0.4306	0.0249	17.2774	0.000
5 p.m.	LOUTFLOW	0.4029	0.0243	16.5636	0.000
	LPRICE	0.5625	0.0235	23.9344	0.000
	LWIND	0.3239	0.0241	13.4636	0.000
	LCONS	0.7717	0.0257	30.0091	0.000
	LINFLOW	0.4358	0.0245	17.7595	0.000
6 p.m.	LOUTFLOW	0.4053	0.0241	16.8422	0.000
	LPRICE	0.5092	0.0242	21.0627	0.000
	LWIND	0.3261	0.0246	13.2381	0.000
	LCONS	0.7802	0.0255	30.6152	0.000
	LINFLOW	0.4407	0.0256	17.2417	0.000
7 p.m.	LOUTFLOW	0.3833	0.0245	15.6669	0.000
	LPRICE	0.4333	0.0235	18.4218	0.000
	LWIND	0.3158	0.0247	12.7842	0.000
	LCONS	0.7760	0.0248	31.2399	0.000
	LINFLOW	0.4453	0.0257	17.3184	0.000
8 p.m.	LOUTFLOW	0.4102	0.0241	17.0076	0.000
	LPRICE	0.3953	0.0236	16.7413	0.000
	LWIND	0.3124	0.0237	13.1789	0.000
	LCONS	0.7808	0.0254	30.7800	0.000
	LINFLOW	0.4169	0.0239	17.4733	0.000
9 p.m.	LOUTFLOW	0.4011	0.0235	17.0450	0.000
	LPRICE	0.4135	0.0260	15.9256	0.000
	LWIND	0.3068	0.0234	13.0990	0.000
	LCONS	0.7760	0.0245	31.6255	0.000
	LINFLOW	0.4117	0.0249	16.5616	0.000
10 p.m.	LOUTFLOW	0.4106	0.0251	16.3412	0.000
	LPRICE	0.4505	0.0252	17.8514	0.000
	LWIND	0.3014	0.0235	12.8028	0.000
	LCONS	0.7718	0.0248	31.1064	0.000
	LINFLOW	0.3627	0.0245	14.7893	0.000
11 p.m.	LOUTFLOW	0.4105	0.0251	16.3726	0.000
	LPRICE	0.4102	0.0226	18.1218	0.000
	LWIND	0.2995	0.0238	12.5609	0.000
	LCONS	0.7804	0.0245	31.9184	0.000
	LINFLOW	0.2465	0.0241	14.3757	0.000
12 p.m.	LOUTFLOW	0.4207	0.0232	18.1599	0.000
	LPRICE	0.4307	0.0236	18.2380	0.000
	LWIND	0.2752	0.0237	11.5864	0.000
	LCONS	0.7848	0.0251	31.2296	0.000
	LINFLOW	0.3660	0.0239	15.2919	0.000
	LOUTFLOW	0.3955	0.0230	17.2135	0.000

Notes: "Ords." stands for order values; "Est. d." is the estimated d ; "Std. Err." denotes the standard error, t represents the t statistic and $P > |t|$ is the p -value of t statistic.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2021.105465>.

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